

DEPARTMENT OF MATHEMATICS

IIT Hyderabad

MA1110 - Calculus - I (B.Tech)

Date: 06-09-2025

Max. Marks 50

Time: 2 hours

(Fractal - part I)

Question Paper

1. Define the following concepts and give two examples of each (no proof required).

[5x2]

- (i) Least upper bound (supremum) of a non-empty subset of $\mathbb{R}$
- (ii) Convergent sequence
- (iii) ϵ - δ definition of continuity of a function at a point
- (iv) Convergent series
- (v) Differentiable function on an interval J

2. For each question, write the correct option in your answer sheet and then justify your answer as a continuation in the same place.

[5x4]

(i)

$$\lim_{n \rightarrow \infty} \frac{(-1)^n(4n+2)}{2n+3}$$

- (A) 4 (B) 2 (C) Does not exist (D) -2

- (ii) Let $h > 0$ is a fixed number. Then

$$\lim_{n \rightarrow \infty} \frac{n^3}{(1+h)^n}$$

- (A) 1 (B) 0 (C) Does not exist (D) 3

(iii)

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

- (A) Convergent series and converges to 3 (B) Convergent series and converges to 1
(C) Divergent series (D) Convergent series and converges to 2

- (iv) Let

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

be a function defined by $f(x) = \sin(x)$, $x \in \mathbb{R}$.

- (A) f is continuous only at $x = 0$ (B) f is continuous only at $x = \pi$
(C) f is no-where continuous on $\mathbb{R}$ (D) f is continuous on $\mathbb{R}$

- (v) Let

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

be a function defined by $f(x) = |x - 2025|$, $x \in \mathbb{R}$.

- (A) f is differentiable only at $x = 2025$ (B) f is differentiable only on $\mathbb{R} \setminus \{0, 2025\}$
(C) f is differentiable only on $\mathbb{R} \setminus \{2025\}$ (D) f is differentiable on $\mathbb{R}$

3. Answer the following:

- (i) Let $\{x_n\}$ be a sequence in $\mathbb{R}$. Show that $\{x_n\}$ is a convergent sequence in $\mathbb{R}$ if and only if $\{x_n\}$ is a Cauchy sequence in $\mathbb{R}$.
- (ii) Let $\{x_n\}$ be a sequence such that $|x_{n+1} - x_n| \leq c|x_n - x_{n-1}| \forall n \in \mathbb{N}$, for some constant c with $0 < c < 1$. Show that $\{x_n\}$ is convergent sequence in $\mathbb{R}$.
- (iii) Let $c > 0$ and $d > 0$. Suppose $\{a_n\}$ and $\{b_n\}$ are sequences of nonnegative real numbers ($a_n \geq 0, b_n \geq 0$ for all $n \in \mathbb{N}$). Assume that

$$ca_n \leq b_n \leq da_n \quad \text{for all } n \in \mathbb{N}.$$

Show that the series $\sum_{n=1}^{\infty} a_n$ converges in $\mathbb{R}$ if and only if the series $\sum_{n=1}^{\infty} b_n$ converges in $\mathbb{R}$.

- (iv) State and prove the Intermediate Value Theorem.
 - (v) State and prove Rolle's theorem.
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